

NutriX: Vision-Grounded Nutrition Understanding and Constrained Meal Planning for Culturally Diverse Foods

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Abstract

Food recognition from images is an important component of automated dietary assessment, but practical nutrition applications require visual identification to be connected with nutritional knowledge, dietary constraints, and meal planning. We present NutriX, a vision-grounded nutrition system that integrates food detection, multimodal food identification, nutrition grounding, and constrained meal planning. The visual component employs a YOLOv8-based detector trained on 123 food classes and 56,147 annotated images. Detected food identities are normalized and linked to structured nutritional knowledge, while barcode and optical character recognition provide complementary identification pathways for packaged foods and nutrition labels. NutriX further incorporates a regional food-knowledge layer containing 217 culturally diverse food classes across ten culinary categories, with associated recipe, ingredient, dietary, and nutritional information. Rather than treating this taxonomy as a 217-class visual benchmark, we use it as a knowledge-grounding resource for culturally diverse nutrition understanding. The grounded food representations are subsequently provided to a constrained meal-planning component that incorporates nutritional targets and user-specific requirements. We evaluate the visual component using detection metrics and application-oriented analyses including regional food subsets, semantic label normalization, rotational robustness, and qualitative failure analysis. The resulting system provides a practical pathway from visual food understanding to nutrition-aware dietary planning while highlighting cultural diversity and heterogeneous food labels as important considerations for food-vision applications.

1. Introduction

Food recognition from images has become an important component of intelligent dietary and nutrition applications. However, practical dietary systems require more than iden-

tifying a food category from an image. A useful system must connect visual food recognition with nutritional information, account for serving quantities and dietary constraints, and translate the resulting information into actionable meal choices. This creates a systems-level challenge at the intersection of computer vision, food understanding, nutrition knowledge, and personalized meal planning.

The challenge becomes more pronounced for culturally diverse diets. Food names, preparation styles, ingredients, and nutritional profiles vary substantially across cuisines, while visually similar dishes can correspond to different food identities. Existing food-recognition systems and datasets are also commonly organized around fixed food categories, making it difficult to directly connect visual predictions with regional food knowledge and downstream nutrition applications. A practical food-vision system should therefore consider not only recognition accuracy, but also how visual predictions can be grounded in structured nutritional knowledge.

We present NutriX, a vision-grounded nutrition system that connects food recognition with nutritional knowledge and constrained meal planning. The core visual component uses a YOLOv8-based food detector trained on 123 food classes and 56,147 annotated images. The detected food identities are normalized and linked to nutritional information, allowing recognized foods to serve as inputs to downstream nutrition analysis and meal planning. NutriX additionally provides complementary food-identification pathways based on barcode and optical character recognition (OCR), enabling the system to handle packaged foods and text-based nutritional information when visual recognition alone is insufficient.

Beyond the visual recognition component, NutriX incorporates a culturally diverse regional food knowledge layer containing 217 food classes across ten culinary categories. The regional repository provides nutritional, recipe, ingredient, and dietary information associated with these food classes. Importantly, this regional taxonomy is treated as a nutrition and food-knowledge resource rather than as a

217-class visual recognition benchmark, since an independent image dataset for these classes is not currently available. This separation allows the visual recognition system and the regional nutrition knowledge layer to be evaluated according to their respective roles while maintaining a unified downstream application.

The resulting pipeline follows a vision-to-nutrition workflow. A food image or complementary identification signal is first converted into a canonical food representation. The representation is then grounded against structured nutritional knowledge, including nutrient values, serving information, ingredient information, and dietary attributes. Finally, the resulting nutritional representation can be provided to a constrained meal-planning component, where nutritional targets and user preferences are incorporated into the selection of meal combinations.

The main contributions of this work are:

- We present a vision-grounded nutrition pipeline that connects food detection with structured nutritional knowledge and downstream meal planning, extending food recognition beyond isolated visual classification.
- We develop and evaluate a 123-class YOLOv8-based food detection system trained on 56,147 annotated images and analyze its behavior using application-oriented evaluation protocols.
- We introduce a culturally diverse food-knowledge layer containing 217 food classes across ten culinary categories, providing regional recipe, ingredient, nutritional, and dietary information for nutrition grounding.
- We investigate practical food-recognition challenges arising from heterogeneous labels and imaging conditions through semantic label normalization, regional food subsets, rotational robustness evaluation, and qualitative error analysis.
- We integrate grounded nutritional information with constrained meal planning, providing a system-level pathway from visual food understanding to nutrition-aware dietary planning.

The remainder of the paper describes the NutriX framework, including its visual recognition and multimodal identification pathways, nutrition knowledge grounding, regional food-knowledge layer, and constrained meal planning. We then describe the experimental setup and evaluate the system through overall detection, regional subset, semantic normalization, rotational robustness, qualitative, and downstream optimization analyses, followed by a discussion of system limitations and future directions.

2. Related Work

2.1. Food Recognition and Dietary Assessment

Food recognition has been studied through image classification, fine-grained recognition, and object detection. Food-

101 established a large-scale benchmark with 101 food categories and 101,000 images, providing an important foundation for food-image recognition research [2]. More recent benchmarks have incorporated nutrition-oriented taxonomies and application-specific food categories [13]. Nutrition5k further connects visual food understanding with nutritional estimation by pairing food imagery with component weights and nutritional annotations [14].

Object detection is particularly relevant to meal images because multiple food items may occur within the same scene. Detection-based approaches localize individual food regions while assigning semantic categories, providing a natural interface to downstream food-level processing. NutriX follows this formulation using a YOLOv8-based detector [8]. However, visual recognition alone does not provide the nutritional context required for dietary applications; recognized foods must be associated with canonical identities, nutritional attributes, and serving information.

2.2. Multimodal Food Identification

Visual recognition is not always sufficient for practical food and nutrition applications. Packaged foods may contain information that is more reliably obtained from product identifiers or textual labels. Food-product databases such as Open Food Facts and FoodRepo provide structured product information that can complement image-based identification [9, 12].

NutriX incorporates barcode and optical character recognition (OCR) pathways around a common food-identity representation. These modalities are used as complementary evidence sources when visual evidence is unavailable, ambiguous, or insufficient, rather than as independent prediction tasks. Existing nutrition-oriented systems have similarly explored combinations of visual recognition, nutritional analysis, and personalized recommendations [5, 16].

2.3. Food and Nutrition Knowledge

Food-recognition systems can be connected to structured nutritional resources to transform semantic predictions into measurable dietary attributes. Resources such as USDA FoodData Central and the Indian Food Composition Tables provide structured nutrient information for grounding food identities in nutritional attributes [6, 15]. Recipe and ingredient resources add information about preparation, composition, and serving context; RecipeNLG and large-scale recipe preference datasets have also supported recipe understanding and personalized food recommendation [1, 11].

Recommendation systems have incorporated nutritional and health constraints through knowledge graphs, personalized meal planning, and constrained recipe selection [3, 7, 10, 17, 18]. NutriX similarly maintains a structured representation connecting food identities with nutritional attributes, ingredients, recipes, dietary classifications, and

serving information.

For culturally diverse applications, this grounding layer is particularly important because food concepts may appear under different names or related labels across culinary traditions. NutriX therefore separates its visual-recognition taxonomy from a broader regional food-knowledge taxonomy. The regional layer provides nutritional, recipe, ingredient, and dietary information without being treated as a 217-class visual-recognition benchmark.

2.4. Positioning of NutriX

NutriX differs from systems that address only food recognition, nutrition estimation, or meal recommendation. It connects visual food recognition, structured nutritional grounding, culturally diverse food knowledge, and constrained meal planning within a single pipeline. The evaluation consequently considers not only overall detection performance, but also regional performance, semantic normalization, rotational robustness, qualitative failure modes, and downstream optimization behavior.

Unlike approaches centered on generative recommendation, the downstream planning component in NutriX uses deterministic constrained optimization after food identity and nutrition grounding. Google OR-Tools with the SCIP solver is used to select candidate recipes subject to nutritional and user-specific requirements [4]. This provides an explicit interface between grounded food information and meal selection.

3. NutriX Framework

NutriX is a vision-grounded food-understanding and nutrition-planning pipeline that converts food observations into canonical food identities, grounds them in structured nutritional knowledge, and provides the result to downstream dietary planning. The framework combines YOLOv8-based food detection with barcode and OCR pathways, a structured food-knowledge layer, regional food information, and constrained meal planning. An overview is shown in Fig. 1.

3.1. System Overview

NutriX accepts food information through multiple input modalities. For image input, the visual recognition component detects food items and assigns semantic categories. For packaged products or visually ambiguous inputs, barcode decoding and OCR provide complementary evidence for food identification or nutritional information. These pathways are mapped to a common canonical food representation before nutrition processing.

The canonical representation serves as the interface between visual perception and downstream nutrition processing. It associates food entities with attributes such as category, ingredients, cuisine, dietary classification, serving in-



Figure 1. Overview of the NutriX vision-grounded nutrition pipeline. Food images and complementary identification signals are converted into canonical food identities, grounded in structured nutrition and regional food knowledge, and passed to constrained meal planning.

formation, and nutritional records. The resulting representation can support food-level analysis or be passed to meal planning.

3.2. Vision-Based Food Detection

The primary computer-vision component uses a YOLOv8-based object detector [8] trained on a 123-class food taxonomy containing 56,147 annotated images. Object detection allows multiple food items to be localized within a single image, making the formulation suitable for meal and plate photographs containing multiple food regions.

The training data contain heterogeneous food imagery spanning different compositions, viewpoints, backgrounds, and presentations. Training and augmentation details are given in Section 4. Each detected instance is represented by its predicted category and localization information, after which the category is passed to the food-normalization layer for association with nutritional and recipe information.

3.3. Multimodal Food Identification

Visual recognition is complemented by barcode and optical character recognition pathways. Barcode processing primarily supports packaged foods, where a product identifier can provide a direct connection to an external product or nutrition record. OCR provides an additional pathway for extracting food or nutrition information from text labels.

Rather than operating as independent applications, these modalities provide alternative evidence sources for constructing the canonical food representation. This allows the framework to retain a common downstream representation when visual evidence is insufficient.

3.4. Nutrition Knowledge Grounding

After food identity is established, NutriX associates the entity with structured nutritional information, including nutrient values, recipes, ingredients, serving attributes, and di-

etary metadata. Serving and quantity information are retained because nutrient values depend on the corresponding quantity or serving basis. Recipe and ingredient information additionally enables representation of prepared foods and composite meals.

3.5. Regional Food Knowledge

NutriX maintains a separate regional food-knowledge layer containing 217 food classes across ten culinary categories. The layer provides recipe, ingredient, nutritional, and dietary information for culturally diverse foods.

The regional taxonomy is treated as a knowledge resource rather than as a visual-recognition benchmark. Because the current regional repository does not provide an independent image dataset covering all 217 classes, NutriX does not claim a 217-class visual recognition result. Instead, the regional vocabulary extends downstream food and nutrition coverage while the visual detector is evaluated on its 123-class benchmark.

3.6. Constrained Meal Planning

Grounded nutritional information is consumed by a constrained meal-planning component implemented using Google OR-Tools with the SCIP solver [4]. Candidate recipes are selected for meal slots subject to nutritional and user-specific requirements.

Let $\mathcal{R}$ denote the candidate recipe set and $\mathcal{M}$ the set of meal slots. For nutritional attribute n , the resulting daily quantity is

$$N_n = \sum_{m \in \mathcal{M}} \sum_{r \in \mathcal{R}} q_{m,r} v_{r,n}, \quad (1)$$

where $q_{m,r}$ denotes the selected quantity of recipe r in meal slot m , and $v_{r,n}$ is its contribution to nutritional attribute n .

The planner supports target ranges for calories and macronutrients together with user-specific restrictions. Additional constraints can represent dietary preferences, allergies, meal-slot requirements, and recipe diversity. This provides a deterministic interface between grounded food representations and feasible meal combinations.

4. Experimental Setup

We evaluate NutriX from the perspective of its primary computer-vision component and its integration with the downstream nutrition pipeline. The evaluation is designed to measure overall food-detection performance while also examining practical issues that arise in heterogeneous food datasets, including regional food coverage, semantically overlapping labels, and food images captured from different viewpoints.

Table 1. NutriX visual food-recognition dataset.

Split	Images	Classes	Purpose
Train	51,535	123	Model training
Validation	2,484	123	Model selection
Test	2,128	123	Final evaluation
Total	56,147	123	–

4.1. Visual Dataset

The primary NutriX food-recognition dataset contains 56,147 annotated images distributed across 123 food classes. The dataset is used to train and evaluate the YOLOv8-based food detector described in Section 3.2. The current evaluation protocol uses separate training, validation, and test partitions to prevent evaluation images from being used during model optimization.

The dataset contains heterogeneous food imagery and therefore includes variation in food presentation, image composition, viewpoint, background, and the number of visible food items. Such variation is relevant to practical dietary applications because food may be photographed individually, on a plate, or as part of a larger meal scene.

4.2. Regional Food Knowledge Evaluation

The regional component is evaluated separately from the visual dataset. It contains a 217-class food taxonomy spanning ten culinary categories and provides recipe, ingredient, nutritional, and dietary information. Because the regional repository does not currently contain a complete image dataset for these classes, the 217-class taxonomy is not used as a visual recognition benchmark.

Instead, the regional layer is evaluated as a knowledge-grounding resource. Food entities can be associated with their corresponding culinary category, recipe information, ingredients, serving information, nutritional attributes, and dietary classifications. This allows the regional layer to extend the semantic coverage of the downstream nutrition system beyond the fixed 123-class visual detector.

For visual analysis of regional foods, we use only food classes for which an appropriate mapping to the existing visual dataset can be established. The resulting regional subset is treated as an analysis of the existing detector rather than as a separately trained 217-class regional detector.

4.3. Training Configuration

The visual recognition component is based on YOLOv8 and is trained using the NutriX 123-class food taxonomy. Training uses the dataset configuration supplied with the project and the corresponding training script. The model is evaluated using the validation and test partitions without modifying the evaluation images during inference.

The training process uses image augmentation to im-

prove robustness to changes in food orientation and presentation. In particular, rotation-aware augmentation is relevant to food images captured from different camera orientations and top-down plate views. Because the available trained checkpoint already incorporates this augmentation, its contribution is not evaluated through a separate retrained baseline in this study. Instead, rotational robustness is evaluated post-hoc by applying orthogonal image transformations to a fixed validation subset, as described in Section 4.6.

4.4. Evaluation Metrics

We evaluate food detection using standard object-detection metrics. Precision measures the proportion of predicted detections that are correct, while recall measures the proportion of ground-truth instances successfully detected. We report mean Average Precision at an intersection-over-union threshold of 0.50, denoted as mAP@50, together with mAP@50:95, which averages average precision over IoU thresholds from 0.50 to 0.95.

For a predicted bounding box B_p and ground-truth bounding box B_g , intersection-over-union is defined as

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}. \quad (2)$$

Precision and recall are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (3)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively.

4.5. Semantic Label Normalization

Food datasets assembled from heterogeneous sources can contain multiple labels referring to the same or closely related food concepts. Such redundancy can introduce ambiguity during evaluation even when the underlying visual food identity is correctly recognized.

We therefore perform a post-hoc semantic label normalization analysis using the existing trained detector. Two groups of potentially overlapping labels are examined: *brinjal/eggplant* and *capsicum/bell pepper*. The original predictions and ground-truth labels are evaluated first using their original class identities and are then mapped to canonical semantic groups.

This analysis does not modify the training dataset or retrain the detector. Its purpose is to determine how much of the observed class-level error can be attributed to semantic label fragmentation rather than purely visual recognition failure.

4.6. Rotational Robustness Evaluation

Food images may be captured under different camera orientations, particularly in mobile dietary logging and top-down meal photography. We therefore evaluate the rotational robustness of the final trained detector without retraining the model.

A reproducible subset of 200 validation images is transformed to four orthogonal orientations: 0° , 90° , 180° , and 270° . Ground-truth bounding boxes are transformed consistently with each image. The same trained checkpoint is evaluated on all four conditions using precision, recall, and mAP@50.

This experiment measures robustness to input rotation rather than the causal contribution of rotational augmentation during training.

4.7. Implementation Details

The NutriX backend is implemented as a service-oriented system around the food-recognition and nutrition-processing components. The application integrates the computer-vision model with food-identity normalization, nutritional data sources, recipe information, and downstream meal-planning functionality.

The meal-planning component uses constrained optimization to select feasible combinations of recipes according to nutritional and user-specific requirements. The computer-vision evaluation is performed independently of the downstream user interface so that the reported recognition metrics measure the visual model itself rather than end-to-end application behavior.

Unless otherwise stated, all quantitative results reported in the following section are obtained from the corresponding project evaluation artifacts. No unmeasured application claims are used as computer-vision performance results.

5. Results

We evaluate NutriX from two complementary perspectives: visual food recognition and downstream nutrition-aware meal planning. The visual evaluation uses the final YOLOv8n checkpoint trained on the 123-class food taxonomy, while the downstream evaluation uses the implemented OR-Tools/SCIP optimization engine on the regional recipe database. In addition to aggregate detection performance, we analyze class representation, culturally relevant Indian food categories, semantic label ambiguity, rotational robustness, and qualitative detection behavior.

5.1. Overall Food Detection Performance

The final YOLOv8n detector achieves a precision of 71.08%, recall of 51.32%, mAP@50 of 58.78%, and mAP@50:95 of 40.56% on the evaluated 123-class benchmark. The difference between precision and recall indicates

Table 2. Overall performance of the NutriX food detector on the 123-class evaluation benchmark.

Metric	Score
Precision	71.08%
Recall	51.32%
mAP@50	58.78%
mAP@50:95	40.56%

Table 3. Class representation across the NutriX dataset splits.

Split	Min	Max	Mean	Median
Train	0	5,014	767.95	165
Validation	0	164	21.41	3
Test	0	141	19.44	7

Table 4. Comparison between the complete visual benchmark and the Indian food subset.

Metric	123-Class	Indian Subset
Classes	123	31
Precision	71.08%	78.94%
Recall	51.32%	74.70%
mAP@50	58.78%	81.56%
mAP@50:95	40.56%	63.33%

that missed instances remain an important source of error.

The final model uses 640×640 inputs with rotation-aware and other spatial augmentations; the best validation mAP@50 was 59.32% at epoch 27.

5.2. Class Distribution and Evaluation Coverage

The validation distribution is highly skewed across the 123-class taxonomy. Although the complete dataset contains 56,147 images, the validation split contains 2,484 images and 6,109 annotated instances. The mean number of validation images per class is 21.41, whereas the median is only 3.

Thirty-seven classes have no validation images, another 30 contain fewer than five images, and only 51 classes contain more than 20 images. As shown in Fig. 2(a), this uneven representation limits the reliability of aggregate metrics as uniform estimates across all categories and motivates class-aware interpretation of the results.

5.3. Indian Food Subset Evaluation

To examine culturally relevant foods, we evaluated the existing 123-class detector on a subset of 31 Indian dish, condiment, and staple-food classes without retraining the model. The subset achieves 78.94% precision, 74.70% recall, 81.56% mAP@50, and 63.33% mAP@50:95, compared with 71.08%, 51.32%, 58.78%, and 40.56%, respectively, on the complete benchmark.

The absolute mAP@50 difference between the two eval-

Table 5. Semantic label normalization for visually overlapping food categories.

Semantic Group	Precision	Recall	F1
Brinjal/Eggplant	92.07%	71.41%	80.44%
Capsicum/Bell Pepper	88.77%	86.27%	87.50%

Table 6. Rotational robustness of the existing NutriX detector on 200 validation images.

Rotation	Precision	Recall	mAP@50
0°	87.10%	84.32%	89.24%
90°	86.04%	83.43%	89.11%
180°	88.08%	82.68%	88.75%
270°	87.06%	81.57%	88.04%

uation settings is 22.78 percentage points. This result indicates stronger recognition performance on the evaluated Indian-food subset than on the complete 123-class benchmark. The difference should be interpreted in the context of the subset’s class composition and representation rather than as evidence that Indian foods are inherently easier to recognize.

Several individual Indian food categories achieve high AP@50, including jalebi (99.10%), onion pakoda (99.10%), palak paneer (96.40%), poha (96.50%), and chole (93.40%). Lower-performing categories include dosa (67.90%), fish curry (63.10%), and shahipaneer (69.40%). One category, rahar ko daal, has no successful detection in the single validation image available for that class. These results reinforce the importance of considering class-level representation when interpreting subset performance.

5.4. Semantic Label Normalization

The dataset contains semantically overlapping labels that can represent the same or closely related visual food concepts. We analyze two such groups: *brinjal/eggplant* and *capsicum/bell pepper*. The analysis uses the existing detector without retraining and applies canonical semantic grouping only during evaluation.

For the brinjal/eggplant group, the original *brinjal* category has zero recall, while the detector recognizes the corresponding *eggplant* category with 71.41% recall. After semantic normalization, the combined identity obtains an F1 score of 80.44%. Similarly, the capsicum/bell-pepper group obtains an F1 score of 87.50% after normalization.

These observations indicate that part of the apparent class-level recognition error is attributable to taxonomy fragmentation. In particular, visually similar or synonymous food names can produce errors that are penalized as class confusions even when the underlying food concept is recognized correctly at a semantic level.

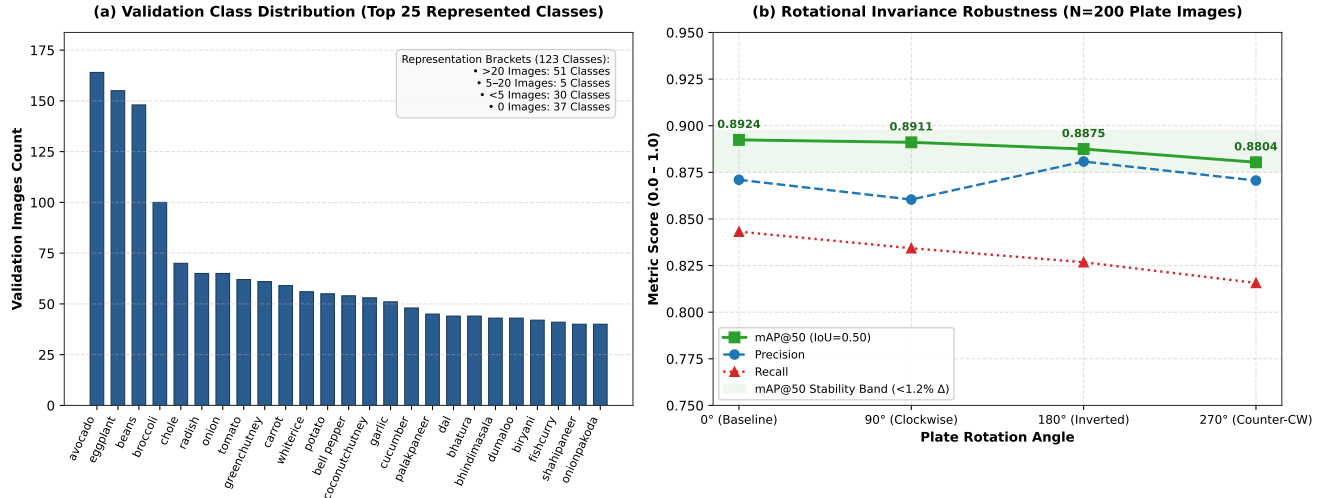


Figure 2. Quantitative evaluation of the NutriX visual component. (a) Validation-set class representation for the 123-class detector, showing the 25 most represented classes together with the overall representation brackets. (b) Rotational robustness on 200 validation plate images evaluated at 0°, 90°, 180°, and 270°.

5.5. Rotational Robustness

We evaluate the existing detector on a reproducible subset of 200 validation images under four orthogonal orientations: 0°, 90°, 180°, and 270°, with ground-truth bounding boxes transformed consistently with each image.

The mAP@50 varies from 89.24% at the original orientation to 88.04% at 270°, a maximum absolute difference of 1.20 percentage points. Precision and recall show similarly small changes across the evaluated orientations, indicating strong robustness under the tested transformations.

This experiment should be interpreted as a robustness evaluation rather than a causal augmentation ablation, since all orientations are evaluated using the same trained checkpoint. Nevertheless, the observed stability is consistent with the use of rotation-aware augmentation during model training.

5.6. Qualitative Error Analysis

We further examine representative validation predictions to characterize practical failure modes. Figure 3 shows seven cases covering correct detection, background false positives, missed detections, visually similar foods, semantic-label confusion, crowded multi-object scenes, and low-confidence detections.

The examples show that errors arise from background objects, visual similarity, sparse class representation, and taxonomy fragmentation. For example, a *capsicum/bell pepper* prediction achieves an IoU of 0.912 despite a mismatch between the ground-truth and predicted class names. These observations are consistent with the quantitative class-distribution and semantic-normalization analyses.

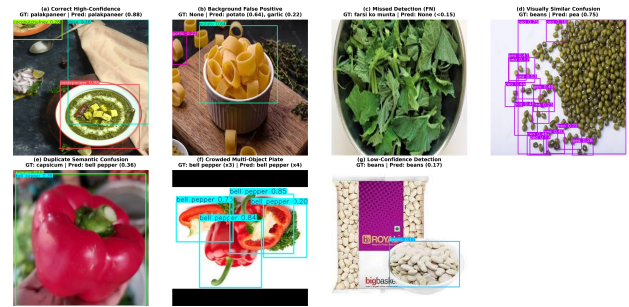


Figure 3. Qualitative food-perception and error analysis across representative validation examples. The examples illustrate (a) correct high-confidence detection, (b) background false positive, (c) missed detection, (d) visually similar confusion, (e) semantic-label confusion, (f) crowded multi-object detection, and (g) low-confidence detection.

5.7. Meal Optimization Benchmark

The downstream meal-planning component was evaluated independently using 80 optimization instances constructed from the 9,139 recipes available in the nutrition database. Twenty instances were evaluated for each candidate recipe-pool size of $N \in \{50, 100, 300, 500\}$. The optimization engine uses OR-Tools with the SCIP solver and considers four meal slots: breakfast, lunch, dinner, and snack.

Mean solve time increases as the candidate recipe pool grows, from 39.25 ms at $N = 50$ to 89.23 ms at $N = 500$, while feasibility remains between 40.0% and 50.0% across the evaluated pool sizes.

The solver uses adaptive nutritional tolerance relaxation, beginning at $\tau = 0.15$ and increasing in increments of 0.05

Table 7. Meal-planning optimization benchmark over 20 instances per candidate pool size.

Pool	Feas.	Mean (ms)	Cal. Err.	Prot. Err.
50	50.0%	39.25	21.06%	27.70%
100	40.0%	43.86	22.23%	28.56%
300	40.0%	81.47	22.74%	28.41%
500	40.0%	89.23	22.74%	28.41%

Parameter	Value
Initial tolerance	15%
Tolerance increment	5%
Maximum tolerance	40%
Initial feasibility	25.0%
Final feasibility	42.5%

up to a maximum of 0.40 when an instance is infeasible. Across all 80 instances, 25.0% were feasible at the initial tolerance, increasing to 42.5% after relaxation. The remaining instances remained infeasible at the maximum tolerance, indicating that simultaneous nutritional, regional, and meal-slot constraints can substantially restrict the feasible solution space.

Overall, the benchmark demonstrates that grounded food representations can be consumed by a constrained meal-planning mechanism. The measured feasibility rates also show that practical dietary planning remains a constrained optimization problem rather than a universally feasible recommendation task.

6. Discussion

The results show that food-recognition performance is strongly influenced by dataset representation and taxonomy design. The 123-class detector achieves 58.78% mAP@50 overall, while the evaluated Indian-food subset reaches 81.56%. This difference should be interpreted in the context of class composition and representation, as the validation distribution has a median of only three images per class and 37 classes have no validation images. The semantic normalization results further show that heterogeneous food labels can inflate apparent class-level errors, with canonical grouping producing F1 scores of 80.44% for brinjal/eggplant and 87.50% for capsicum/bell pepper.

The rotational evaluation indicates that the existing detector remains stable under the tested orthogonal transformations, with mAP@50 changing by at most 1.20 percentage points. This provides evidence of robustness to the evaluated image rotations, although it does not establish the causal contribution of rotation-aware augmentation. At the systems level, the meal-planning benchmark shows that grounded food representations can be consumed by deterministic constrained optimization. Increasing the candidate recipe pool from 50 to 500 raises mean solve time from 39.25 ms to 89.23 ms, while adaptive tolerance relaxation

increases feasibility from 25.0% to 42.5% across the 80 evaluated instances.

Several limitations remain. The 217-class regional food layer is a knowledge-grounding resource rather than an independently evaluated visual-recognition benchmark, and the visual dataset is substantially imbalanced across classes. The Indian-food subset therefore should not be interpreted as evidence of universally superior recognition for Indian foods. Similarly, the rotational experiment evaluates a fixed checkpoint rather than a controlled augmentation ablation, and the meal-planning benchmark measures computational feasibility and latency rather than clinical dietary outcomes or long-term user benefit. Accordingly, NutriX should be viewed as a vision-grounded nutrition-processing framework that connects food recognition, structured nutritional knowledge, culturally diverse food representation, and constrained planning, rather than as a clinically validated dietary decision system.

7. Conclusion

We presented NutriX, a vision-grounded nutrition framework that connects food detection with structured nutritional knowledge, culturally diverse food representation, and constrained meal planning. The 123-class YOLOv8-based detector achieves 58.78% mAP@50 overall and 81.56% mAP@50 on the evaluated Indian-food subset, while semantic normalization demonstrates that part of the apparent class-level error can arise from heterogeneous food taxonomies. The rotational evaluation further shows stable detection under the tested orthogonal transformations.

Beyond visual recognition, NutriX provides a structured pathway for mapping recognized food identities to nutritional and regional knowledge and subsequently using these representations in constrained meal planning. The downstream optimization experiments demonstrate practical computational latency while also exposing the feasibility limitations created by simultaneous nutritional and meal-specific constraints. Overall, the results support the use of vision-grounded food representations as an interface between computer vision and nutrition-oriented applications, while highlighting dataset representation, semantic consistency, and culturally diverse food coverage as important directions for future food-vision research.

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